

Convex Optimization 作业 2

October 2025

1 Solve following LP by Simplex Tableau

$$\begin{aligned} & \max_{x,y} 3x + 2y \\ & \text{s.t.} \quad \begin{cases} 2x + y \leq 18 \\ 2x + 3y \leq 42 \\ 3x + y \leq 24 \\ x, y \geq 0 \end{cases} \\ & \text{s.t.} \quad \begin{cases} z - 3x - 2y = 0 \\ 2x + y + s_1 = 18 \\ 2x + 3y + s_2 = 42 \\ 3x + y + s_3 = 24 \\ x, y, s_1, s_2, s_3 \geq 0 \end{cases} \end{aligned}$$

	z	x	y	s_1	s_2	s_3	
	1	-3	-2	0	0	0	0
s_1	0	2	1	1	0	0	18
s_2	0	2	3	0	1	0	42
s_3	0	3	1	0	0	1	24



$$\begin{aligned}\textcircled{0} &= \textcircled{0} + \textcircled{3}, \\ \textcircled{1} &= \textcircled{1} - \textcircled{3} \cdot \frac{2}{3}, \\ \textcircled{2} &= \textcircled{2} - \textcircled{3} \cdot \frac{2}{3},\end{aligned}$$

	z	x	y	s_1	s_2	s_3	
	1	0	-1	0	0	1	24
s_1	0	0	$\frac{1}{3}$	1	0	$-\frac{2}{3}$	2
s_2	0	0	$\frac{7}{3}$	0	1	$-\frac{2}{3}$	26
x	0	3	1	0	0	1	24



	z	x	y	s_1	s_2	s_3	
	1	0	0	3	0	-1	30
y	0	0	$\frac{1}{3}$	1	0	$-\frac{2}{3}$	2
s_2	0	0	0	-7	1	4	12
x	0	3	0	-3	0	3	18



	z	x	y	s_1	s_2	s_3	
	1	0	0	$\frac{5}{4}$	$\frac{1}{4}$	0	33
y	0	0	$\frac{1}{3}$	$-\frac{7}{6}$	$\frac{1}{6}$	0	4
s_3	0	0	0	-7	1	4	12
x	0	3	0	$\frac{9}{4}$	$-\frac{3}{4}$	0	9

$$s_1 = 0, s_2 = 0, s_3 = 3, x = 3, y = 12, z = 33$$

2 Solve following LP by Simplex Tableau

$$\begin{array}{ll} \max. & -x_1 + 3x_2 + x_3 \\ \text{s. t.} & \begin{cases} 3x_1 - x_2 + 2x_3 \leq 7 \\ -2x_1 + 4x_2 \leq 12 \\ -4x_1 + 3x_2 + 8x_3 \leq 10 \\ x_1, x_2, x_3 \geq 0 \end{cases} \end{array}$$

$$\text{s. t.} \begin{cases} z + x_1 - 3x_2 - x_3 = 0 \\ 3x_1 - x_2 + 2x_3 + s_1 = 7 \\ -2x_1 + 4x_2 + s_2 = 12 \\ -4x_1 + 3x_2 + 8x_3 + s_3 = 10 \\ x_1, x_2, x_3, s_1, s_2, s_3 \geq 0 \end{cases}$$

	z	x_1	x_2	x_3	s_1	s_2	s_3	
	1	1	-3	-1	0	0	0	0
s_1	0	3	-1	2	1	0	0	7
s_2	0	-2	4	0	0	1	0	12
s_3	0	-4	3	8	0	0	1	10



	z	x_1	x_2	x_3	s_1	s_2	s_3	
	1	$-\frac{1}{2}$	0	-1	0	$\frac{3}{4}$	0	9
s_1	0	$\frac{5}{2}$	0	2	1	$\frac{1}{4}$	0	10
x_2	0	-2	4	0	0	1	0	12
s_3	0	$-\frac{5}{2}$	0	8	0	$\frac{3}{4}$	1	1



	z	x_1	x_2	x_3	s_1	s_2	s_3	
	1	0	0	$-\frac{3}{5}$	$\frac{1}{5}$	$\frac{4}{5}$	0	11
x_1	0	$\frac{5}{2}$	0	2	1	$\frac{1}{4}$	0	10
x_2	0	0	4	$\frac{8}{5}$	$\frac{4}{5}$	$\frac{6}{5}$	0	20
s_3	0	0	0	10	1	1	1	11



	z	x_1	x_2	x_3	s_1	s_2	s_3	
	1	0	0	0	$\frac{13}{50}$	$\frac{43}{50}$	$\frac{3}{50}$	$\frac{583}{50}$
x_1	0	$\frac{5}{2}$	0	0	$\frac{4}{5}$	$\frac{1}{20}$	$-\frac{1}{5}$	$\frac{39}{5}$
x_2	0	0	4	0	$\frac{16}{25}$	$-\frac{2}{5}$	$-\frac{4}{25}$	$\frac{456}{25}$
x_3	0	0	0	10	1	1	1	11

$$s_1 = 0, s_2 = 0, s_3 = 0, x_1 = \frac{78}{25} = 3.12, x_2 = \frac{114}{25} = 4.56, x_3 = \frac{11}{10} = 1.1, z = \frac{583}{50} = 11.66$$

3 Solve following LP by Simplex Tableau

$$\begin{aligned} \max. \quad & -3x_1 + 5x_2 + 2x_3 + x_4 \\ \text{s. t.} \quad & \begin{cases} x_1 + x_2 + x_3 \leq 4 \\ 4x_1 - x_2 + x_3 + 2x_4 \leq 12 \\ -x_1 + x_2 + 2x_3 + 3x_4 \leq 12 \\ x_1, x_2, x_3, x_4 \geq 0 \end{cases} \\ & \begin{cases} z + 3x_1 - 5x_2 - 2x_3 - x_4 = 0 \\ x_1 + x_2 + x_3 + s_1 = 4 \\ 4x_1 - x_2 + x_3 + 2x_4 + s_2 = 12 \\ -x_1 + x_2 + 2x_3 + 3x_4 + s_3 = 12 \\ x_1, x_2, x_3, x_4, s_1, s_2, s_3 \geq 0 \end{cases} \end{aligned}$$

	z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	
	1	3	-5	-2	-1	0	0	0	0
s_1	0	1	1	1	0	1	0	0	4
s_2	0	4	-1	1	2	0	1	0	12
s_3	0	-1	1	2	3	0	0	1	12



	z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	
	1	8	0	3	-1	5	0	0	20
x_2	0	1	1	1	0	1	0	0	4
s_2	0	5	0	2	2	1	1	0	16
x_4	0	-2	0	1	3	-1	0	1	8



	z	x_1	x_2	x_3	x_4	s_1	s_2	s_3	
	1	$\frac{22}{3}$	0	$\frac{10}{3}$	0	$\frac{14}{3}$	0	$\frac{1}{3}$	$\frac{68}{3}$
x_2	0	1	1	1	0	1	0	0	$\frac{4}{32}$
s_2	0	$\frac{-19}{3}$	0	$\frac{4}{3}$	0	$\frac{5}{3}$	1	$\frac{-2}{3}$	$\frac{3}{3}$
x_4	0	$\frac{-2}{3}$	0	$\frac{1}{3}$	1	$\frac{-1}{3}$	0	$\frac{1}{3}$	$\frac{8}{3}$

$$s_1 = 0, s_2 = \frac{32}{3}, s_3 = 0, x_1 = 0, x_2 = 4, x_3 = 0, x_4 = \frac{8}{3} = 2.6667, z = \frac{68}{3} = 22.6667$$